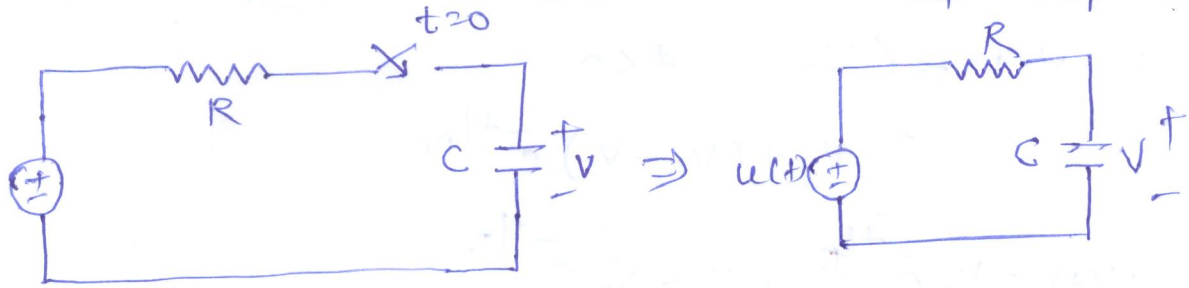
Step Response of an RC ckt :-

When the dc source of an RC circuit is suddenly applied the voltage or current source can be modeled as a step function the response is known as a step response.



As voltage does not change instantaneously $V(0^-) = V(0^+) = V_0$,
where V_0 = initial voltage across capacitor

$V(0^-)$ → Voltage across the capacitor just before switching

$V(0^+)$ → Voltage across capacitor just immediately after switching.

By KVL

$$V_s - I_c R - V_c = 0$$

$$V_s\, u(t) = RC \frac{dv_c}{dt} + V_c$$

$$\frac{dV_c(t)}{dt} = \frac{V_s\, u(t) - V_c}{RC}$$

where V = voltage across capacitor

for $t \geq 0$, $u(t) = 1$

so $$\frac{dV}{dt} = -\frac{1}{RC}(V - V_s)$$

$$\Rightarrow \int \frac{dV}{V - V_s} = \int -\frac{1}{RC}\, dt$$

Integrating both side, Introducing initial condition, we have

$$\left[\ln (V - V_s)\right]_{V_0}^{V(t)} = \left[\frac{-t}{RC}\right]_0^t$$

$$\ln (V(t) - V_s) - \ln (V(0) - V_s) = -\frac{t}{RC}$$

$$\ln \left[\frac{V(t) - V_s}{V(0) - V_s}\right] = -\frac{t}{RC}$$

$$\Rightarrow \frac{V(t)-V_s}{V(0)-V_s} = e^{\frac{-t}{RC}}$$

$$\Rightarrow V(t)-V_s = (V(0)-V_s)\, e^{-t/RC}$$

$$\Rightarrow \boxed{V(t) = V_s + (V(0)-V_s)\, e^{-t/RC}}$$

thus $$V(t) = \begin{cases} V_0, & t<0 \\ V_s + (V_0 - V_s)\, e^{-t/RC} \end{cases}$$

so $V(t) = V_0 e^{-t/\tau} + V_s - V_s e^{-t/\tau}$

$V_0 e^{-t/\tau}$ ↓ natural response (V_n)

$V_s - V_s e^{-t/\tau}$ ↓ forced response V_f

$V = V_n + V_f$

$V_n = V_0 e^{-t/\tau}$ → natural response

$V_f = V_s(1-e^{-t/\tau})$ → forced response due applied external voltage source V_s

With increase of time natural response diesout and also in forced response ~~due to applied~~ the steady state kept and other also dies out. so we can break complete response into two

Complete response = Temporary + steady state

= Transient response + steady state response

so $V = V_t + V_{ss}$

$V_t = (V_0 - V_s)\, e^{-t/\tau}$

$V_{ss} = V_s$

The Transient response V_t is temporary: It is the portion of the complete response that decays to zero as time approaches infinity

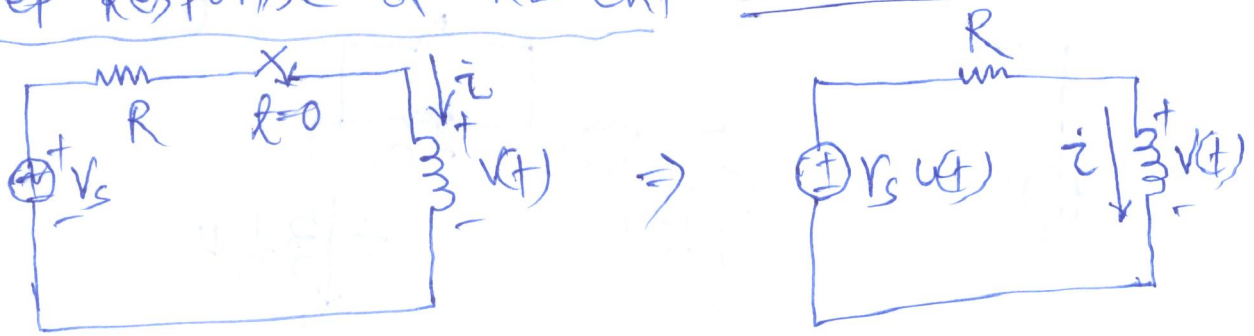
so from two above deco we know that the natural response and transient response are same and also forced response and steady state response so

$$\boxed{V(t) = V(\infty) + [V(0) - V(\infty)]\, e^{-t/\tau}}$$

where $v(0)$ = initial voltage at $t = 0^+$

$v(\infty)$ = final or steady state voltage

Step Response of RL CKT Lect-12



Total Response

$i(t) = i_t + i_{ss}$ = transient + steady state

but we know that $i_t = Ae^{-t/\tau}$ decaying in nature and $i_{ss} = \frac{V_s}{R}$

So $i(t) = Ae^{-t/\tau} + \frac{V_s}{R}$

at $t = 0$

$i(0) = A + \frac{V_s}{R}$

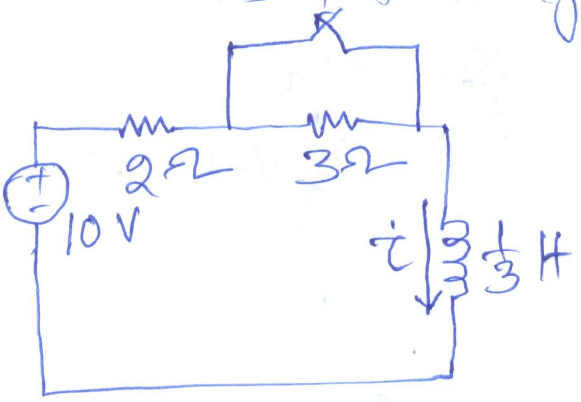
$\Rightarrow A = i(0) - \frac{V_s}{R}$

$\Rightarrow i(t) \left(i(0) - \frac{V_s}{R}\right) e^{-t/\tau} + \frac{V_s}{R}$

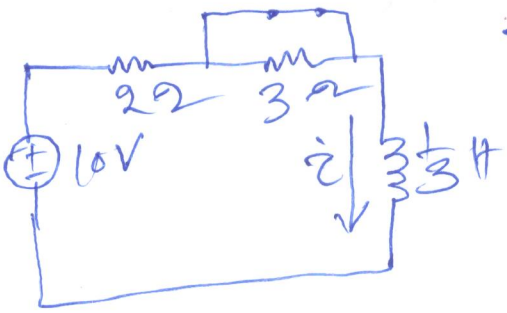
$$i(t) = i(\infty) + \left(i(0) - i(\infty)\right) e^{-t/\tau}$$

Pb: Find $i(t)$ in the ckt for $t>0$, Assume that the switch has been closed for long time



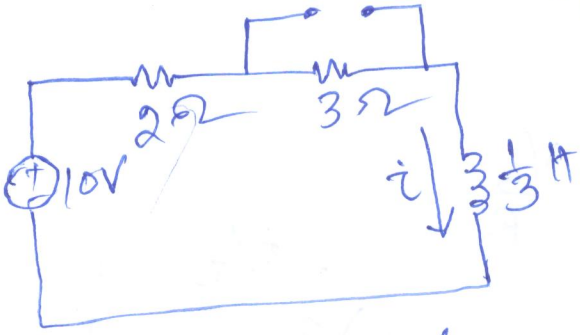


Ans: for $t<0$,



$$i(0) = \frac{10}{2} = 5\ A$$

for $t>0$,



$$i(\infty) = \frac{10}{5} = 2\ A$$

$R_{th} = 5\ \Omega$

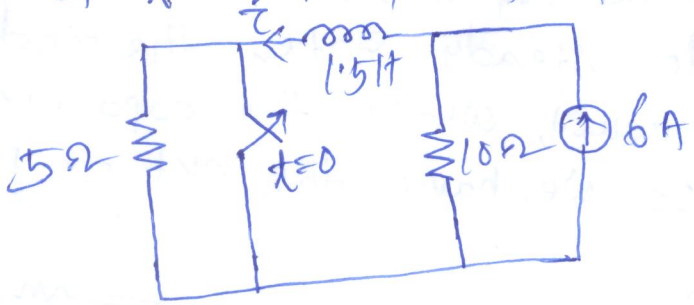
$$\tau = L/R = \frac{1/3}{5} = \frac{1}{15}\ sec$$

So $i(t) = i(\infty) + (i(0) - i(\infty))\, e^{-t/\frac{1}{15}}$

$= 2 + (5-2)\, e^{-15t}$

$$\boxed{i(t) = 2 + 3e^{-15t}}\ Amp,\ t>0$$

Pb:- The switch has been closed for long time. It opens at $t=0$. Find $i(t)$ for $t>0$.



Ans:- for $t<0$, $i(0) = 6A$ as switch is closed

for $t>0$, switch is open. So

$$i(\infty) = \frac{10\times 6}{15} = 4\ A$$

$R_{th} = 15\ \Omega$

$$\tau = \frac{15\times 15}{30} = \frac{45}{2}$$

$$\tau = \frac{L}{R} = \frac{1.5}{15} = \frac{15}{10\times 15} = \frac{1}{10}\ sec$$

$$i(t) = i(\infty) + (i(0) - i(\infty))\, e^{-t/\tau}$$

$$= 4 + (6-4)e^{-10t}$$

$$\boxed{i(t) = 4 + 2e^{-10t},\ t>0}$$

Pb:- The switch in fig. 8.2 has been closed for a long time. It is open at $t=0$. Find

(a) $i(0^+)$, $v(0^+)$

(b) $\frac{di(0^+)}{dt}$, $\frac{dv(0^+)}{dt}$

(c) $i(\infty)$, $v(\infty)$

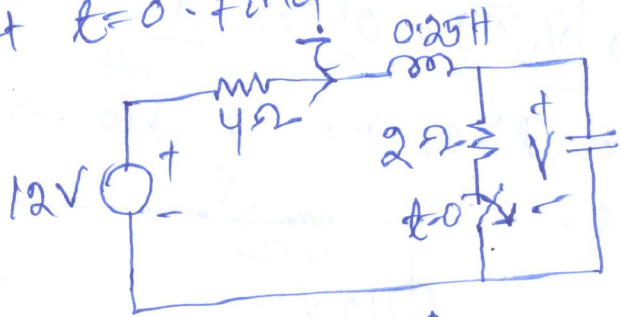


Fig. 8.2

DC power flow:-

Ans:-

(a) If the switch is closed a long time before $t=0$, it means that the ckt has reached dc steady state at $t=0$. At dc steady state, the inductor acts like a short circuit, while the capacitor acts like an open ckt, so we have the ckt in Fig. 8.3(a) at $t=0^-$. Thus

$$i(0^-) = \frac{12}{4+2} = 2A, \quad V(0^-) = 2\, i(0^-) = 4V$$

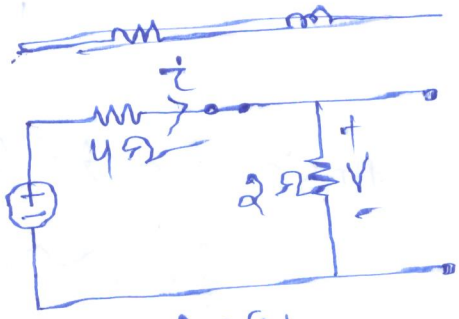
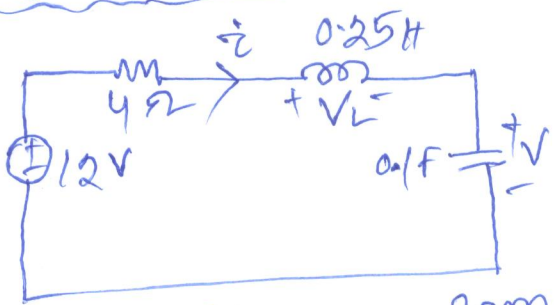


Fig-8.3(a)

(b) At $t=0^+$, the sw is open:-



$i(0^+) = i(0^-) = 2A$, same current will the capacit

$V(0^+) = V(0^-) = 4V$ i.e $i_c(0^+) = 2A$

$$\Rightarrow c\frac{dV_c(0^+)}{dt} = i_c(0^+)$$

$$\Rightarrow \frac{dV_c(0^+)}{dt} = \frac{i(0^+)}{c} = \frac{2}{0.1} = 20\ V/s$$

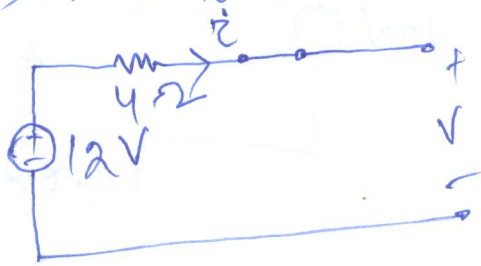
Similarly $L\frac{di}{dt} = V_L \Rightarrow \frac{di}{dt} = V_L/L$

Now from the KVL, we have

$$12 - 4\times i(0^+) - V_L(0^+) - V(0^+) = 0$$

$$\Rightarrow \boxed{V_L(0^+) = 0} \Rightarrow \frac{di(0^+)}{dt} = \frac{V_L(0^+)}{L} = \frac{0}{0.25} = 0\ A/s$$

(c) For $t>0$, i.e $t \to \infty$ → ckt reaches steady state.



$i(\infty) = 0$

$V(\infty) = 12V$